

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0422

Toa Vaca 138 kV
January 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0422. The Interconnection Customer proposes a 98.2 MW solar photovoltaic facility interconnecting at the Toa Vaca 138 kV substation in the Virginia area. The request was received on January 11, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 175 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,886,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Toa Vaca - Thomaston 138 kV	Chaffee Gas Recovery - Blaine 138 kV line	184.4	159	36.4
Stonewall Energy Center - Blaine 138 kV	Thomaston - Fountain 138 kV	127.6	230	6.7
Fountain - Pleasant Vly 138 kV	Luna - Yavapai 138 kV	125.5	266	16.4
Blaine - Chaffee Gas Recovery 138 kV	Fountain - Thomaston 138 kV	123.7	264	6.7
Stonewall Energy Center - Fountain 138 kV	Fountain - Thomaston 138 kV	117.5	223	6.7
Pleasant Valley - Cub-Run Dp 138 kV	Luna - Yavapai Substation 138 kV L	113.1	278	16.4
Luna - Yavapai 138 kV	Pleasant Vly - Fountain 138 kV	109.2	246	11.9
Lake Crk - Sierra Pacific Lumber 138 kV	Taber - Glenannan Substation 345 kV	106.9	782	5.1
Bunker Hill - Beltsville 138 kV line	Taber - Glenannan 345 kV	105.1	812	5.4
Rocky Reach - Idaho Springs 138 kV	Sideburn - Yavapai 138 kV	104.3	255	6.9
Fountain - Thomaston Substation 138 kV	Chaffee Gas Recovery - Blaine 138 kV Line	104.2	316	15.2
Nasa - Bunker Hill 138 kV line	Glenannan - Taber Substation 345 kV	104.2	855	5.4
Fairland - Siuc 138 kV L	Taber Substation - Glenannan 345 kV	103.3	643	5.3
Luna - Peavy Fall Hy 138 kV	Pleasant Valley - Fountain 138 kV	102.4	228	11.9
Burke-Sideburn - Rocky Reach 138 kV	Sideburn - Yavapai 138 kV	101.9	238	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Toa Vaca - Thomaston 138 kV	Rebuild line with higher-capacity conductor	\$2,262,000	29% (\$651,000)
Stonewall Energy Center - Blaine 138 kV	Rebuild line with higher-capacity conductor	\$1,524,000	100% (\$1,524,000)
Fountain - Pleasant Vly 138 kV	Rebuild line with higher-capacity conductor	\$5,526,000	100% (\$5,526,000)
Blaine - Chaffee Gas Recovery 138 kV L	Rebuild line with higher-capacity conductor	\$2,049,000	100% (\$2,049,000)
Stonewall Energy Center - Fountain 138 kV	Rebuild line with higher-capacity conductor	\$1,150,000	100% (\$1,150,000)
Pleasant Valley - Cub-Run Dp 138 kV line	Rebuild line with higher-capacity conductor	\$2,668,000	100% (\$2,668,000)
Luna - Yavapai 138 kV line	Reconductor line	\$951,000	100% (\$951,000)
Lake Ck - Sierra Pacific Lumber 138 kV line	Reconductor line	\$497,000	1% (\$5,000)
Bunker Hill Substation - Beltsville 138 kV	Reconductor line	\$212,000	1% (\$2,000)
Rocky Reach - Idaho Springs 138 kV	Reconductor line	\$607,000	100% (\$607,000)
Thomaston - Foutain 138 kV	Reconductor line	\$468,000	100% (\$468,000)
Nasa - Bunker Hill 138 kV line	Reconductor line	\$836,000	1% (\$7,000)
Fairland - Siuc 138 kV L	Reconductor line	\$547,000	2% (\$12,000)
Luna - Peavy Fall Hy 138 kV	Reconductor line	\$901,000	100% (\$901,000)
Burke-Sideburn - Rocky Reach 138 kV	Reconductor line	\$365,000	100% (\$365,000)
	Total allocated to project		\$16,886,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.